# CHAPTER 3

## The First Biological Psychiatry

**F**or two centuries, the predominant theme in psychiatry has been harnessing the neurosciences to therapeutics. In the asylum, nineteenth-century psychiatry ran into a dead end in achieving this goal. But outside the asylum as well, in what one might call "the first biological psychiatry," nineteenth-century alienists attempted to enlist the neurosciences in caring for patients. As with the asylum, this initial attempt to give psychiatry a scientific basis came to a dead end. What began as a promising attempt to limn the biological and genetic roots of psychiatric illness ended in the specter of "degeneration," the notion that inherited mental illness worsened steadily over the generations. Yet the parallel to the story of the asylum is striking. In neither the asylum nor biological psychiatry did the failure of the idea mean necessarily that the idea itself was faulty. Rather, it was that after 1900 the whole paradigm for looking at illness changed.

### *Enter Ideas*

The first biological psychiatry was a movement of ideas rather than an exercise in bricks and mortar. It asked, How do the genetics and the chemistry of the brain actually make people ill? What therapies might succeed with such illnesses? The answers were not to be found in the humdrum routine of asylum life but in research done in universities and institutes. What made the first biological psychiatry distinctive from previous humoral theories was not the belief that psychiatric illness possessed an underlying neural structure—physicians since the Ancients have believed that—but the desire to lay bare the relationship between mind and brain through systematic research. It is this systematic element that was new, doing

experiments on humans and animals, testing drugs, and examining brains postmortem.

Research in psychiatry was part of a larger nineteenth-century current toward research in medicine in general. Doctors began to apply the clinical-pathological method: reasoning back and forth from findings at autopsy to the signs and symptoms the patient displayed before death. This cross-reasoning permitted researchers to identify distinctive diseases. If one were interested in lung disease, for example, one would attempt to link the gurglings and rattlings in the lungs of a living patient to the findings at autopsy: Different diseases, such as emphysema and pneumonia, clearly separable at autopsy, made different noises antemortem. Psychiatry in the nineteenth century too tried to attempt this clinical-pathological method, hoping to demonstrate the correctness of the biological approach rather than merely accept it on faith. Generally speaking, these efforts took place not in asylums but in universities.

Thus there are two narratives to keep track of, although they largely overlap. One is the research narrative: How did biological psychiatry press forward? The other is the teaching narrative: How did psychiatry in general become a university subject? Linking both of these narratives is the practical reality that the family doctor had to know something about psychiatric illness to deal with upset families and mentally ill patients. The whole subject had to be de-demonized for the average medical student, who arrived in class with the same preconceptions as the lay public. Family physicians had to learn that mental illness was a familiar aspect of medical experience, as opposed to a satanic curse visited on the unfortunate by dark powers. In their practices, they would encounter much in the way of mania and depression, panic disorders, and dementia, and the stakes were enormous in deciding who was sick, who needed to be treated at home, and who required institutional care.

Sensitizing family doctors meant that mental illness must be “medicalized,” or drawn into the circle of medical experience. Medicalization in psychiatry occurred at the same time as other physicians medicalized diseases such as tuberculosis and nephritis. The underlying logic was the same: Just as family doctors had to know something of the lung to treat TB, they needed to be familiar with the brain and central nervous system to treat mental illness. “Is it not necessary,” asked Parisian psychiatrist Ernest Billod rhetorically in 1884, “for the physician to possess knowledge of the illness that is as complete as possible.... in the grave eventuality of placing a patient

in an institution?" The consequences of making a mistake could be dreadful he pointed out, "stigmatizing" the family with insanity.[1] The doctor had to know what he was doing.

Therefore in the nineteenth century, university instruction in psychiatry became organized all across the Atlantic community. To teach medical students, first there had to be lectures, then departments of psychiatry—or "clinics" as they are called in Europe—with access to psychiatric beds, and ultimately big psychiatric research institutes. The first biological psychiatry was thus driven simultaneously by the demands of education and scientific curiosity.

Within the very first reformed asylums of the eighteenth century alienists had begun lecturing to medical students. In 1753, William Battie started lecturing in psychiatry at St. Luke's Hospital in London, the governors of which had permitted Battie to attract "more gentlemen of the faculty ... to make this branch of physic their particular care and study."[2] In 1805, Vincenzio Chiarugi commenced lecturing at his asylum in Florence.[3]

These were interesting beginnings. But for the next hundred years teaching and research in psychiatry would be dominated by the Germans.

## *A German Century*

Giving so much attention to one country, or one language, in an international history of psychiatry requires some justification. The warrant is that the Germans truly did dominate the field. In modern medicine, the great majority of eponyms, or things named after individuals, are named after Germans. As one opens a medical dictionary to "signs," pathological phenomena that the physician discovers in a patient, the page swims with German names: "Berger's sign," a deformation of the pupils in early neurosyphilis, named after the Austrian ophthamologist Emil Berger; Möbius's sign, the inability to keep the eyeballs converged in thyroid disease, named after the Leipzig psychiatrist Paul Julius Mobius (who assured himself notoriety in the history of medicine by writing in 1900 a book titled *The Physiological Feeble-Mindedness of Women*); and Westphal's sign, the loss of the knee-jerk reflex in nervous disease.[4] This German dominance was owing to the many state-supported universities and asylums at which, from

1800 on, research and teaching were proceeding apace.

There were two mechanisms for hooking teaching to research in the German university. One was the tradition of the doctoral dissertation, for all medical students had to produce a dissertation to get the MD degree (the French did too, but there were many more German students). Second was a postdoctoral research project called the "habilitation" required from PhDs and MDs aspiring to teach at universities. Unlike the dissertation, often a scientific puffball, the habilitation had to make a serious contribution to knowledge to be accepted. After the habilitation, one would first be a private docent, entitled to lecture, then an associate professor (*ausserordentlicher Professor*), then a full professor (*ordentlicher Professor*). In no other country were so many students and postgraduates harnessed in this manner to publication and research. This academic structure, plus ample funds from the ministry, guaranteed scientific preeminence to Germany until 1933.

In Central Europe, lecturing in psychiatry began with Johann Heinroth in 1811 in Leipzig. Why is Ernst Horn, appointed in 1806 associate director of the Charité Hospital in Berlin and having responsibility for the psychiatry unit in that institution, not mentioned in first place? Because his formal psychiatry teaching there did not begin until 1818; and only after Karl Wilhelm Ideler took over psychiatry at the Charité in 1832 did it become an autonomous department.[5] None of these early professors fit the later model of chair-holder simultaneously in charge of an academic department and a psychiatric hospital (or "clinic"). Heinroth had no access to psychiatric patients. Indeed, the discipline vanished from Leipzig with his death in 1843 (not to resume there again until the appointment of Paul Flechsig in 1877, a tunnel-visioned brain anatomist).[6] Before 1865, such sporadic initiatives were undertaken at other German universities as well, such as Würzburg (1834) and Munich (1861).

Such was the strength of asylum psychiatry over university teaching in those days that a powerful asylum man such as Christian F. W. Roller could cause the newly founded (1826) teaching department at Heidelberg to collapse, as in 1842 Roller transferred the Heidelberg patients to his own new asylum at Illenau.[7]

These early efforts usually failed to teach the medical students much about psychiatry because the lectures themselves were situated in asylums often far distant from the students' other classrooms. Demonstrating the patients to the students was frequently less than illuminating because the asylums housed

mostly chronic cases whose symptoms all seemed more or less alike. And the asylum psychiatrists themselves were often too preoccupied with administration to show much interest in teaching.[8] In their rustic isolation, they were also cut off from medicine. As one university psychiatrist later put it, “In no branch of medicine were there so many peculiar eccentrics as among the older alienists.”[9]

What was needed was a department of psychiatry situated close to the other medical departments, one that could admit patients whose illnesses the professors wanted to demonstrate in lectures and that could make psychiatry a convincing part of medicine by belonging to a general hospital. This was achieved in Germany in 1865, when Wilhelm Griesinger, then 48, whose career had flitted back and forth from internal medicine to psychiatry, agreed to become the professor of psychiatry in Berlin at the Charité hospital. Not only was Griesinger to become the single most influential representative of the first biological psychiatry, he established the modern model of the department of psychiatry as dedicated to teaching and research rather than to custodialism. It was with Griesinger that university psychiatry triumphed over asylum psychiatry.

Born in 1817 in Stuttgart, Griesinger had studied medicine in nearby Tübingen, then in Zurich under Johann Schonlein, returning to Tübingen for his degree. Schönlein was the man most responsible for turning German medicine away from philosophical considerations about the nature of life and toward the natural sciences. This was a time when questions were just starting to be asked about the biological substrate of illness in the tissues of the body, and the techniques of chemistry, physiology, and microscopy deployed to answer them. Under Schönlein, Griesinger came to understand that a physician should be trained as a scientist, studying patients through direct observation at the bedside rather than relying on thousand-year-old observations about humors.

Wilhelm Griesinger, professor of psychiatry in Berlin 1865-1868, who is considered to be the founder of “the first biological psychiatry” (courtesy of National Library of Medicine).

Shortly after graduating from Tübingen in 1838, Griesinger spent two years as an assistant physician at the newly opened Winnenthal asylum near Stuttgart, which was then under the direction of a dynamic young psychiatrist named Albert Zeller, a member of the generation of reformers who had taken their inspiration from Jacobi at Siegburg. At the end of his Winnenthal stay, Griesinger permitted himself an astonishing piece of hubris for a 28-year-old doctor: He published in 1845 a textbook of psychiatry, which indeed achieved some recognition.[10]

At this point, Griesinger abandoned psychiatry, becoming an assistant physician at the internal medicine clinic in Tübingen. Then he moved on to

such way stations as running the outpatient department at Kiel in North Germany, becoming the personal physician of the vice-regent of Cairo (relying on these experiences to publish a big book on infectious illnesses). Griesinger returned to Tübingen in 1854—at great remove from psychiatry—as the professor of medicine. Finally in 1860, he took on the professorship of internal medicine in Zurich, where he had begun almost 30 years previously as a medical student. It is actually remarkable that the bulk of the career of this the most famous German psychiatrist before Kraepelin should have been spent in quite another field.

Yet over his long trawl through internal medicine, Griesinger never fully lost sight of psychiatry, demonstrating psychiatric cases at his medicine clinic in Tübingen whenever he had the chance, or spontaneously organizing a course of lectures in psychiatry at the Zurich hospital. In 1861, Griesinger had returned to his textbook, completely revising it and bringing it out in a new edition, the text of which was in accordance with his current thinking about psychiatric illness as brain disease, or "nerve disease," as he put it. In contrast to the first edition, this second revised edition became a huge scholarly success and was perhaps the single most influential psychiatry textbook in the Western world until Kraepelin's big book in the 1890s (see p. 191).[11]

When Wilhelm Ideler, the Romantic psychiatrist who had been the previous holder of the psychiatry chair in Berlin, died in 1860, Griesinger was the obvious candidate to replace him. In 1865, Griesinger accepted the headship of psychiatry there. Although he would arrive to discover the psychiatric clinic at the Charité in complete disarray, he found the soil in Berlin already well prepared for his notion that mental disease was brain disease. Berlin was rapidly rivaling Vienna as the world epicenter of medicine in those years, and Griesinger's old teacher Schönlein, for example, had been called to Berlin in 1840 as the professor of pathology and therapy, placing internal medicine there on a scientific footing that emphasized the physical examination of the patient in alliance with laboratory findings. The pathologist Rudolf Virchow, another great Berlin name, had often studied brains postmortem. (Psychiatrist Carl Wernicke later disparaged Virchow's brain sections, saying "This is the way you cut cheese."[12]) Thus in Berlin, Griesinger found a community of physicians accustomed to thinking of themselves as scientists and avid for leadership in the area of psychiatric illness.

Griesinger divided the Charité clinic into halves, one for “the usual nervous diseases,” the other for “nervous disease with a primarily psychiatric presentation.” Alternating between them by semester, he would hold three clinical lectures a week from 7 A.M. to 9 A.M. By the spring semester of 1867, 46 medical students were attending regularly (an impressive number, in view of the fact that psychiatry would not be on the exam), in addition to younger psychiatrists working in asylums, and a contingent of foreign physicians visiting Berlin. The medical students could also take individual patients out into the clinic’s garden and practice testing their reflexes and looking at their eyeballs.[13]

The house staff (physicians in training) at the Charité experienced psychiatry not as it was practiced in an asylum but in a general hospital, for example, cross-covering other services. Robert Wollenberg, at the time a psychiatry resident, recalled being summoned full of fear and trembling to the obstetrical service in order to terminate a prolonged labor. Under the critical eyes of the staff midwives, he ended up performing a difficult maneuver called a “version.” This meant reaching into the uterus, grasping the baby’s feet and extracting it, an operation that would challenge any experienced obstetrician.[14] (Today’s psychiatry residents, by contrast, tremble at the prospect of even doing an neurological exam.)

The Charité residents, as Griesinger told them, were there to learn how to diagnose. It was part of the new spirit that one had to watch an illness unfold for a while before one could determine its character, hence the notion of a short-stay urban asylum that would take on patients with few formalities (and little stigma), then either discharge them well or ship them off to a regular asylum.[15]

In 1867, Griesinger founded the flagship journal of the new neurologically oriented psychiatry, the *Archive for Psychiatry and Nervous Diseases,* which would overshadow the old *Journal of General Psychiatry* of the asylum doctors. The preface to the first issue of Griesinger’s Archive contained one of those programmatic statements that in terms of historical resonance was comparable to Lenin’s *What Is to Be Done?* “Psychiatry has undergone a transformation in its relationship to the rest of medicine,” Griesinger said. “This transformation rests principally on the realization that patients with so-called ‘mental illnesses’ are really individuals with illnesses of the nerves and brain” (“...dass die sogennanten ‘Geisteskranken’ Hirn- und Nerven-kranke Individuen sind”). Psychiatry must therefore “emerge from its closed-off

status as a guild and become an integral part of general medicine accessible to all medical circles."[16] These are among the most portentous words ever uttered in the history of psychiatry. They inaugurated a new era of university psychiatry devoted to the study of brain and nervous tissue. In October 1868, a year after this first issue of the new *Archive* appeared, Griesinger died at age 51 of a ruptured appendix.[17]

Griesinger-style clinics now began appearing in a number of universities. Already in 1866, Ludwig Meyer, who as chief of the Hamburg asylum had made a name for himself by auctioning off the restraints used to bind patients, became head of a newly built university psychiatric clinic attached to an asylum in Göttingen. Meyer was co-editor of Griesinger's Archive and belonged to the inner circle of academic "nerve doctors" (*Nervenärzte)* .[18] Further foundations followed. In 1872 in Munich, the brilliant student of brain structure Bernhard von Gudden transformed the psychiatric clinic there, which had been first opened in 1859, into a major research center. (Gudden died in 1886, dragged to the bottom of Lake Starnberg by his patient, the crazed Bavarian king Ludwig II.)[19] Carl Fürstner established a modest university clinic in Heidelberg in 1878 (the first asylum built within a university explicitly on Griesinger's lines).

Meanwhile in Austria, events unfolded in their own manner and in a way quite independent of Griesinger. In 1848, the 15-year-old Theodor Meynert moved with his family from Dresden, where he had been born, to Vienna, where his father was settling down to work as a journalist (Meynert's mother had originally been Viennese).[20] Meynert studied medicine in Vienna, having as a student the good fortune to be recognized as promising by the great pathologist Carl von Rokitansky. After graduating in 1861, Meynert specialized under Rokitansky in "the architecture and function of the brain and spinal cord." Thus there is no question of Meynert ever having been prepared as a psychiatrist. He was a neuropathologist and as such took a post as pathologist at the Vienna asylum. Yet from 1868 onward, Meynert also began lecturing in psychiatry. In 1870, the Austrian government appointed him associate professor of psychiatry, bringing him immediately into conflict with the older generation of humanitarian, psychologically oriented asylum clinicians. Meynert was a pupil of the Viennese school of natural science and of therapeutic nihilism, which maintained that the treatment of untreatable diseases is useless. He himself believed that he was in psychiatry in order to do research and not necessarily to cure patients. Indeed, he went on to do

pioneer work in the microscopic structure of the brain and spinal cord, a challenging task involving discovering stains—carmine dye in Meynert's case—with which to visualize the various brain cells under the microscope, then differentiating layers (the cytoarchitecture) of the cerebral cortex and other specialized parts of the brain.[21] He worked on the fourth floor of his home, hunched over his microscope, hunting not just for structure but for pathological lesions in the neurons as well. And for this search, Meynert became an object of mockery among generations of the psychoanalytically oriented historians who came later, secure in the knowledge that Sigmund Freud, who had been a pupil of Meynert, had found the right way. That Meynert discovered little aside from the lesions of neurosyphilis did not necessarily mean that he was wrong, merely that he was searching with the wrong tools.

While Meynert's focus on the brain's frontal lobes would survive among subsequent generations as an object of humor, it was forgotten that Meynert, perhaps next to Griesinger the most famous representative of the first biological psychiatry of the nineteenth century, had forecast the second biological psychiatry of the twentieth. As Meynert wrote in 1890: "The study of human anatomy in its current form has passed from a solely descriptive science to something higher, to a form of knowledge that attempts to explain.... The more that psychiatry seeks, and finds, its scientific basis in a deep and finely grained understanding of the anatomical structure [of the brain], the more it elevates itself to the status of a science that deals with causes."[22] These were prophetic words.

Yet Meynert, like so many of his generation, had an Achilles heel when it came to patients. He basically had little interest in them, believing most to be beyond help. The Austrian physician and playwright Arthur Schnitzler, who trained for six months at Meynert's psychiatric clinic at the Vienna General Hospital, gave a sense of the man's clinical style: "[He] was a great scholar, a splendid diagnostician, but as a physician in the narrower sense, in his personal relations with the patients, ... he did not win my admiration. As masterful as he may have been in the face of disease, in front of the sick person his behavior often seemed to me cool, uncertain, if not indeed anxious."[23] It probably was also unhelpful that Meynert was a chronic alcoholic.[24] Malicious tongues claimed Meynert's only connection with psychiatry was that he had gone through a delirium tremens.

Theodor Meynert's slide box, containing sections of various parts of the brain. Meynert tried to find under the microscope the physical causes of mental illness (courtesy Institut fur Geschichte der Medizin, Vienna).

Nonetheless, Meynert was a pioneering figure. In 1868, he called for a fundamental reorientation of psychiatry—quite independently of Griesinger who was making the same demand in Berlin—away from preoccupation with labeling symptoms and toward understanding the underlying anatomical

origins of psychiatric illness.[25] Meynert's work was the beginning of the last phase in the development of the first biological psychiatry: concentration on anatomy.

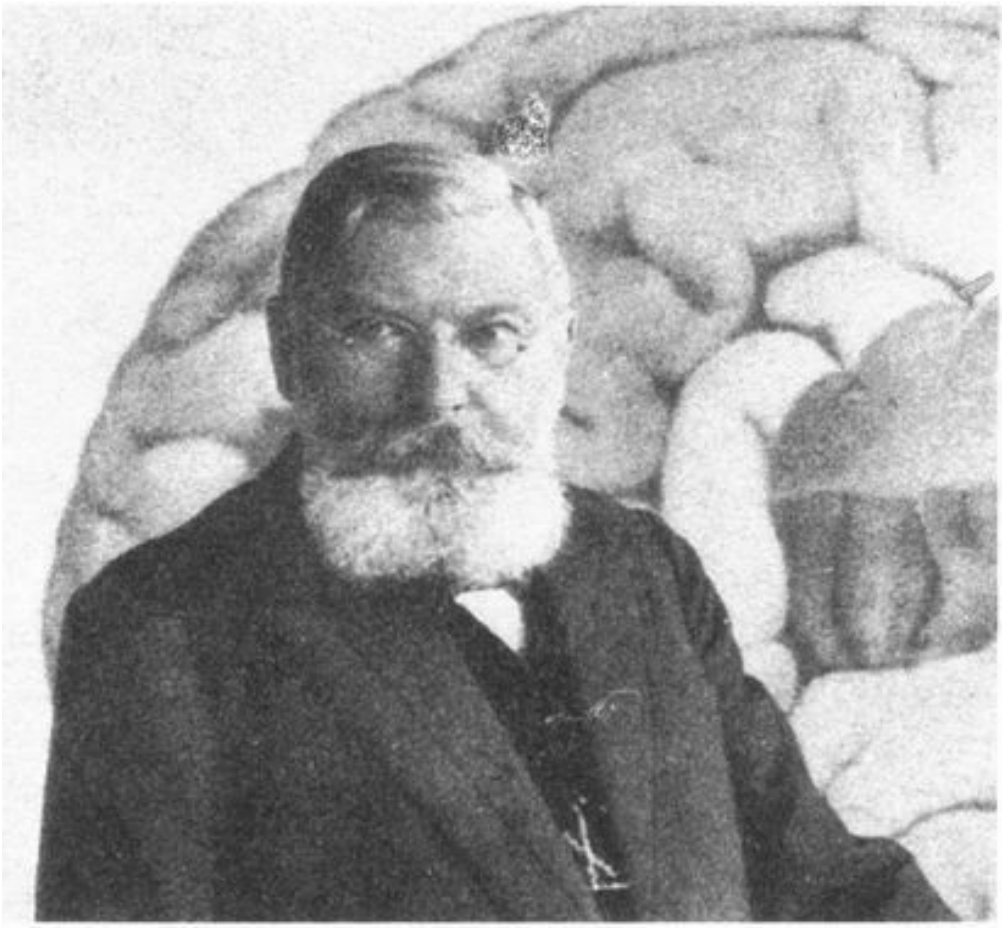
In the 1880s and after, an absolute craze for studying psychiatry with the microscope took possession of the German, Austrian, and Swiss universities. It is generally agreed that this craze led to a dead end, and that the first biological psychiatry died because it detached itself too completely from patients and their world. Yet misguided or not, these researchers were trying to achieve something more than merely mapping the architecture of the grey and white matter of the cerebral cortex. Following the model of neurosyphilis, they were trying to identify specific lesions in patients whose illnesses seemed to be primarily psychiatric rather than neurological. In other words they considered themselves at this point to be psychiatrists rather than neurologists (neurology in many places was a branch of internal medicine). The leaders in this field were almost all students of Meynert.

Two giants in the study of neuroanatomy and cerebral localization opened university clinics in the 1880s: Paul Flechsig in Leipzig in 1882 and Eduard Hitzig in Halle in 1885. Flechsig laid down the basic map of what regions of the cerebral cortex are responsible for what functions ("cerebral localization"), and Hitzig established that the brain responds to electrical stimulation. Both men had immense achievements.[26] Both were also terrible clinicians.

Here biological psychiatry in Germany showed the same shadow side as in Austria. Believing psychiatric diseases to be incurable, many of these professors concentrated on research in the basic sciences and displayed little interest in clinical psychiatry. They had not gone into the discipline to make their patients better. Emil Kraepelin, who worked briefly as a resident under Flechsig until leaving in disgust, recalled Flechsig as being completely uninterested in learning about patients or their problems.[27] For Daniel Paul Schreber, a jurist who apparently had neurosyphilis and spent some time around 1884 in Flechsig's clinic, Flechsig was the "murderer of my soul."[28] (It was Schreber whom Freud in 1911 attempted retrospectively to psychoanalyze.[29]) Hitzig had the general reputation of being "intolerable." He knew little of psychiatry and, in one colleague's judgment, would never have lasted on the staff of a big psychiatric clinic had his scientific accomplishments not placed him in charge of one.[30] Accordingly, as research into brain anatomy and brain physiology accelerated, the first biological

psychiatry became nihilistic about the possibilities of clinical care, seeking instead to become an annex of the basic medical sciences.

Paul Flechsig, professor of psychiatry at Leipzig 1877-1921, who pioneered the study of cerebral localization (but who was poor at dealing with patients) (courtesy of Museen der Stadt Wien).



The first biological psychiatry slid to an end in the work of Carl Wernicke, narrowly balanced between “neuromythology” and brain anatomy. Wernicke, the most ambitious of the neurological psychiatrists, set out to establish that specific symptom complexes could be associated with specific areas of the brain. Here he had tremendous initial success, enshrining his name at the age of 24 in medical history. Having just graduated in 1870 from Breslau, he went to Vienna to work with Meynert. Wernicke established in 1874 that

when a patient has a stroke in a specific part of the brain (the posterior perisylvian region), he or she will be unable to understand the spoken word and will be able to speak only in incomprehensible jargon. The area of the brain became known as Wernicke's area, and this particular aphasia became called "Wernicke's aphasia." This scientific windfall would stamp Wernicke's interests permanently. He spent the next years in Berlin attempting to identify other such areas, publishing between 1881 and 1883 a three-volume book on brain disease.[31]

At this point, Wernicke turned to a possibly more fanciful subject: whether psychiatric symptom-complexes existed that might be localized in certain areas of the brain. In 1885, he became professor of psychiatry at Breslau, spending almost all the rest of his days trying to match symptoms in patients to postulated brain abnormalities.[32] He developed thereby his own vocabulary for the accounting of psychiatric and neurological complaints. Months after his move in 1904 to the psychiatric clinic in Halle, he was mortally wounded when his bicycle was run over by a truck. Clinician to the last, as he lay dying he said reflectively, "I'm perishing of autopsychic disorientation."[33] Wernicke's vocabulary did not outlive him.

With Wernicke, the development in biological psychiatry that Karl Jaspers later characterized as "brain mythology" reached its end point.[34] Emil Kraepelin would shortly announce that the course of psychiatric illness offered the sharpest clue to its nature, rather than, as Wernicke believed, the kind of symptoms the patient had at any particular moment. Kraepelin espoused a longitudinal approach, Wernicke a cross-sectional. It was the historic victory of Kraepelin's views over Wernicke's that marked the end of the first biological psychiatry.

By 1911, Germany had 16 Griesinger-style university psychiatric clinics, 187 public asylums, and 225 private asylums, not to mention the many additional institutions in Austria and German-speaking Switzerland. By that year, Germany had almost 1,400 physicians specializing in psychiatry. [35] This pool of research energy was unmatched anywhere else in the world and explains the primacy of Central Europe in the science of mind and brain. But just as the particular development of the German states had given science policy a certain stamp, events in other countries would stamp the neurosciences there just as ineffably.

## *French Disasters*

Unlike Germany, the history of university psychiatry in France is not synonymous with the biological approach, for Esquirol in particular had little interest in the integuments of the brain. What is intriguing about the French story is how politics destroyed virtually all efforts to launch university psychiatry of any kind—neuroscientific or psychosocial-before a government decree in 1877 finally created psychiatric clinics at the four main universities.

At some point before 1815, Pinel had offered a clinic in psychiatry for students at the Salpêtrière in addition to his university lectures in internal medicine and pathology. The clinic did not survive. In 1817, Esquirol began teaching a private course in psychiatry at the Salpêtrière. It was not an official course of the faculty, and not until 1821 did Antoine Royer-Collard begin offering the faculty's psychiatry lectures. They ceased a year later as the government abolished the entire School of Medicine in Paris for political reasons.[36] Therewith, formal teaching in psychiatry in Paris came to an end for the next half century.

France had gifted psychiatrists, many of whom made lasting contributions to the discipline. Yet their energies were never harnessed in the systematic manner of the German system with its 20-odd universities and armies of private-docents and professors.[37] The experience of the brilliant young Antoine-Laurent Bayle is instructive. In a doctoral thesis in medicine in 1822, Bayle attributed the psychiatric symptoms of neurosyphilis to a chronic inflammation of the meninges, thus making himself the first to discover a psychiatric disease with definite organicity, establishing that as the underlying disease worsens, the symptoms worsen as well.[38] What was the fate of this French Wernicke? Did he become the professor of psychiatry at Lyons (an analogous example to Wernicke's professorship at Breslau)? Hardly. Owing to political intrigues, Bayle never received even an asylum post to say nothing of a university professorship. Bayle was associated politically with Royer-Collard, under whom he had studied at Charenton. Royer-Collard had a falling out with the Pinel-Esquirol circle, and just after Royer-Collard died in November 1825, the members of this circle forced Bayle to take a position as assistant librarian in the Paris faculty of medicine. After returning to asylum psychiatry a few years later, his appetite for discovery presumably quenched, Bayle made no significant further

contributions.[39]

Such maneuvers occurred regularly in France. Another victim was Josef Babinski, the gifted psychiatrist and discoverer of the reflex named after him (raking the sole of the foot will cause the big toe to turn up if brain disease, an upper-motor neuron lesion, is present). Political rivalries in the 1890s prevented Babinski from ever gaining an academic post.[40] A highly centralized system was permitted to shoot itself in the foot, so to speak, because everything was controlled from a single ministerial desk. And because opportunities in the provinces were so scant, there was no alternative venue to which people such as Bayle or Babinski could retreat, unlike the German psychiatry professors who regularly drifted from chair to chair. Wernicke, for example, had gone to Halle in 1904 after the Breslau city council interfered with the running of his service.

Thus in France, politics decreed that research in psychiatry would be situated in the asylum and not in the university. Of these asylum-based scholars, best known, or perhaps most notorious, was Bénédict-Auguste Morel, among the first to describe schizophrenia. It was Morel who launched the notion of “degeneration” on its fateful trajectory. Born into a family that became impoverished at the death of the father, Morel attended a religious boarding school in eastern France. Expelled from school, he came to Paris and became something of a penniless bohemian, toying with journalism and sharing a flat with the young Claude Bernard (who later became a famous physiologist but at the time, like Morel, was poverty-stricken). Both Morel and Bernard decided for medicine. In 1839, Morel submitted a doctoral dissertation that touched among other things on insanity. Resolving to go further into psychiatry, he asked Bernard (then an interne at the Salpêtrière) to introduce him to Jean-Pierre Falret, psychiatrist and chief physician of the hospice. Thus Morel became involved in clinical psychiatry. In 1841, accompanying a patient on a voyage abroad, Morel chanced on an asylum for mentally retarded children run by Dr. Johann Jakob Guggenbühl at Abendberg, Switzerland. Guggenbühl specialized in children suffering from cretinism (later determined to be the result of iodine deficiency). The Abendberg visit hooked Morel’s interest in the whole subject of mental retardation. Given the organicity of several forms of this condition such as Down syndrome, Morel would find it an easy step to organicist assumptions about psychiatry in general.

In 1848, Morel became staff physician at the Maréville asylum near

Nancy, where he strode forth as a great reformer, releasing the patients from their fortresslike cells. Three years later, in 1851, he began teaching a course in psychiatry there. Although Morel's views might on occasion possess a romantic flavor, a result of his deep piety, there is no doubt that he was firmly within the biological tradition. "I believe that the brain is the organ of the soul," he declared.[41] By 1856, when Morel became chief physician of the St.-Yon asylum, he had acquired a kind of *idée fixe* about "degeneration" in its various forms, including mental retardation. Accordingly, Morel, the first major figure in France to have lectured in psychiatry since Esquirol and Royer-Collard, lined up on the biological side.[42]

To teach within the French university system, one has to pass a difficult examination called the *agrégation.* (Another reason why French psychiatry was behind: Young scientists spent all their time memorizing facts rather than doing research.) Morel did not become *agrégé.* But another of the comrades who hung about with Morel and Claude Bernard in their student days did pass the *agrégation:* Charles Lasègue.

It was Lasègue who rekindled the university teaching of psychiatry in Paris (as opposed to teaching non-academic courses in hospitals in the 1840s and after; several clinicians—such as Jules Baillarger at the Salpetriere—offered hospital courses). Seven years younger than Morel, Lasègue had launched his career by the same route: He asked Claude Bernard to introduce him to Falret. Like Morel, Lasègue became captivated by psychiatry. Both men completed their MD dissertations at the age of 30. But rather than going off to some peripheral mental hospital as Morel did, upon graduation in 1846 Lasègue remained in Paris at the forensic clinic of the Prefecture of Police. He passed his *agrègation* in 1853 and became a staff physician in the Paris hospitals. Finally in 1862, he was given permission to teach a course in psychiatry at the faculty of medicine—yet without access to patients. Lasègue inaugurated the course on November 28 of that year in the Faculty's great, and only, lecture theater. He shifted further lectures to the Necker Hospital.

One might conclude that at last the genius of the French system had triumphed, that Lasègue, a truly talented man who had already made several original contributions to psychiatry (in 1873 he would describe anorexia nervosa for the first time as a distinct illness), had finally reached the point of Griesinger at the Charité in 1865 or Meynert in Vienna in 1870. But no. In 1867, five years after beginning his course, Lasègue became appointed professor of general pathology, not of psychiatry, for which there was as yet

no chair. In 1869, Lasègue received the chair of clinical medicine at Pitié Hospital.[43] Few significant psychiatrists would train under him. Once again, the system had run into the sand.

Now psychiatry and nervous disease were becoming an embarrassment to the Ministry and to the Parliament, where there was great enthusiasm for such teaching. In 1875, two commissions recommended that a full chair in psychiatry ("maladies mentales") be created, and in 1877 a government decree determined that each of France's four medical faculties—the Sorbonne in Paris, and the faculties in Lyons, Nancy, and Marseille—would have teaching in psychiatry at the level of a full professorship. Here once again, the Ministry and Faculty defeated themselves. By this time, several distinguished psychiatrists in Paris were giving hospital-based lectures in psychiatry. Who should become the professor? After much political arm-twisting, Lasègue's candidate prevailed: an obscure 45-year-old psychiatrist named Benjamin Ball, born in Naples of an English father and a Swiss mother. Ball had little going for him save his powerful patron. At Lasègue's instigation, in the early 1870s Ball started to offer a complementary course of lectures in psychiatry at the Faculty. In 1879, he began lecturing at the Sainte-Anne asylum as the new chair-holder. But when the authorities created a chair for Ball at Sainte-Anne, they neglected to make him simultaneously medical director of the asylum. Intramural conflict at Sainte-Anne plagued Ball during his tenure in office. A disappointment to everyone, he died discouraged in 1893 at the age of 58.[44]

Following these negotiations is rewarding only because they illustrate why France remained a second-rate psychiatric power. Few stories, however, demonstrate so perfectly this inability to rise from mediocrity as the story of Jean-Martin Charcot, a man often hailed as a great psychiatrist though in fact he was an internist and pathologist, and understood almost nothing of major psychiatric illness. Born in 1825 of a working-class family, Charcot was a self-made man, a scholar and researcher who by the force of his own wit had pushed his way into a system dominated by upper-middle-class clinicians and bureaucrats. After jumping through the usual academic hoops, in 1862 Charcot became the chief physician of the Salpêtrière, a hospice that had its own medical service in addition to the psychiatric staff of which Charcot was not a member. In 1866, he began lecturing on chronic diseases at the Salpêtrière. Because the hospice, with its large population of aging (and mostly non-insane) women was a sort of treasure trove of pathology, Charcot

became attracted to the study of neurological illness, just as Meynert had discovered the organic side of research in his association with Rokitansky. But Meynert at least taught on a psychiatric service, which Charcot never did. In the 1860s, Charcot made some fundamental discoveries in the area of neuropathology, linking anatomic changes to clinical symptoms in multiple sclerosis, for example, or describing ALS (amyotrophic lateral sclerosis, or "Lou Gehrig's disease"). These were major contributions and earned Charcot such a reputation that by the 1870s he was the best-known physician in France. In 1882, a chair in nervous disease was created specially for him.

In the early 1870s, Charcot became entangled in what would later constitute office-practice psychiatry, the undifferentiated mass of neurotic complaints that he termed "hysteria." Believing hysteria to be a real organic disease, transmitted genetically and associated with presumptive but unidentified changes in nervous tissue, Charcot elaborated a kind of air-castle of "iron laws of hysteria," a disease thought to follow its own rules. Thus it was that from the early 1870s until his death in 1893, Charcot poured most of his energy into hysteria, directing his many students toward it and forming, or deforming, an entire generation of researchers, many of whom were psychiatrists. He possessed titanic authority. All of Europe came to believe in Charcot's "hysteria," which willy-nilly had become the centerpiece of French psychiatry.

But after Charcot's death the whole edifice of Charcotian hysteria collapsed. The supposedly organic nervous disease called "grand hysteria" turned out to be nothing more than an artifact of suggestion, as did the presumed "stigmata" of hysteria, fixed physical signs such as supposed tunnel vision. An entire generation of physicians and patients had been gulled into believing, and reproducing in their illness behavior, the symptoms of a set of iron laws that never existed.[45] It was a complete disaster for French psychiatry, from which the French even by World War II had not fully recovered. (A Parisian symposium in 1925 on the centenary of Charcot's birth makes piteous reading.[46])

This history of misery was for the most part built into the French system. Only in France could a handful of chair-holders in Paris dominate the destiny of medical training in a country of almost 40 million people. When a system like the French produces a Pasteur, its extreme centralization has the potential to mobilize large amounts of talent. When it produces someone like Charcot, quite lacking in common sense and grandiosely sure of his own judgment, it

harbors the potential for calamity.

One more calamity lay in the wings for French psychiatry in the nineteenth century. It took the form of Valentin Magnan, chief physician of the Sainte-Anne asylum.[47] Next to Morel, Magnan was perhaps the second most famous of France's biological psychiatrists. Born in 1835, he came up from Lyons where he had studied medicine to serve a prized internship in Paris, first at Bicêtre in 1864 under the organicist Victor Marcé and the hereditarian Prosper Lucas, then in 1865 for his second year of internship at the Salpêtrière. At the time, the Salpêtrière was a site of "prodigious activity"—as Magnan's biographer put it—in the area of neurology and organically oriented psychiatry.[48] Magnan involved himself fully in the research of such teachers as Jean-Pierre Falret. It was probably from Falret that Magnan derived his lust for classification that would later prove so fateful in the international isolation of French psychiatry.

Just after finishing his thesis in 1866, Magnan won a post in the admissions bureau of the Sainte-Anne asylum, which would open a year later. (The curious Parisian admissions system called for patients to be funneled from the forensic service of the Prefecture of Police to the Sainte-Anne admissions bureau, whence they would be distributed to the various asylums of the Paris region.) Here, Magnan stood in the midst of the great stream of psychiatric pathology coursing through the metropolis: In 1868 alone, he and his colleagues examined 2,600 patients. In that year, Magnan initiated at Sainte-Anne the kind of hospital-based lecture series then being attempted at other asylums and hospitals as well. (The course was shut down in 1873 by the authorities because of a public scandal over the demonstration of patients to students, resuming four years later.[49]) Passed over for the professorship in favor of Ball, Magnan remained at Sainte-Anne, the gray eminence of French psychiatry, until his retirement in 1912.

Among the patients passing through the admissions bureau, Magnan became particularly interested in the alcoholic, the epileptic, and those suffering nerve damage from the then popular drink absinthe (later banned as a neurotoxin). In short, Magnan singled out that class of patients whom Morel had identified as "degenerate," and in his writings Magnan would soon spread the doctrine of "degeneration" far and wide within Europe (see p. 181). That was one disservice to French psychiatry.

Magnan's other disservice was to elaborate, in imitation of Charcot, his own set of iron laws about the supposed evolution of psychotic illness, in

which patients who began with fretfulness would end in dementia, the distinctively French “chronic delusional disorder with progressively downhill course” (*délire chronique à évolution systématique*). With the exception of neurosyphilis and Alzheimer’s disease, no major psychiatric illness behaves in this manner, and the notion of carving out chronic delusional disorders as a privileged category makes little sense, for delusions are found in tandem with other symptoms in almost all psychotic illness. Yet Magnan’s idiosyncratic approach to delusions stuck within French psychiatry and became a national tradition. It blocked the acceptance of schizophrenia, isolating France from the rest of the international community and stunting its growth behind a wall of nationalistic pride.[50]

Magnan represented the high-water mark in France of the first biological psychiatry. It was a version too permeated with his genetic reductionism to survive. After World War I, “mental hygiene” and other catchphrases of the psychosocial version of psychiatry would carry the day.

## *Anglo-Saxon Laggards*

When in 1904 a young American psychiatry resident named Clarence Farrar visited London on a busman’s holiday, he wanted quite naturally to tour the Bethlem Royal Hospital, most famous of all British psychiatric institutions. Staff physician William Stoddart, known as a “vast, portly figure arrayed in morning coat and top hat,” came forward to show young Farrar around. The visitor asked his guide to characterize the state of English psychiatry. “Stoddart says English psychiatry is far behind that on the continent, that there exists in England no school of psychiatry,” Farrar noted in his diary. “In Bethlem there is no pathologist and no pathology done except what Dr. Stoddart ‘finds time’ to do. In histological diagnosis, G.P. [general paralysis] is the only disease he thinks it possible to recognise. No psychology is done. It is considered to be much work with little result.”[51] Farrar, who had just come from the great pathology laboratories of Heidelberg where such chiefs as Franz Nissl and Aloys Alzheimer bent over the microscope night and day to find the cause of psychosis and dementia, could scarcely believe the backward state of English psychiatry.

In England, where everything depended on private and charitable

initiatives and nothing on public funds, research into biological psychiatry scarcely existed. “The following sketch,” said surgeon Walter Rivington in his portrayal of medicine in the United Kingdom in 1879, “will show how widely different are the systems pursued in Paris and in London. In Paris 1 school of medicine, 1 staff of professors, 1 group of laboratories and museums, lecture theatre and dissecting-room.... responsibility to the State and centralisation are the main characteristics. In London, 11 schools, 11 staffs of professors and assistants, with 11 separate courses on each subject, 11 sets of offices and school buildings with their lecture theatres and laboratories.... Appointment of hospital staffs, without competition, by the lay authorities of the hospital through private influence or favour leavened by recommendation of future colleagues....”[52] In this English system of extreme decentralization, no money could be found for laboratories or research institutes and little time spared from clinical duties for the microscope. Stoddart was in fact not untypical in his indifference to peeling back the mysteries of nature with laboratory research.

All this is not to say that British psychiatry was not biologically oriented. From the earliest days, British physicians had assigned a physical seat to mental illness. As Bedlam’s John Haslam wrote in 1809, “From the preceding dissections of insane persons, it may be inferred that madness has always been connected with disease of the brain and its membranes.” [53] From autopsies done at Hanwell, William Charles Ellis gained strength for his theories that “increased sanguineous action takes place on the commencement of insanity.”[54] Yet while a somaticist theme ran throughout British psychiatry in the nineteenth century, it was sharply accented in the academic setting. Virtually without exception, the men who brought psychiatry to Britain’s medical students were biologically oriented.

Since Battie’s time, there had been sporadic lecturing on psychiatry in the London hospital medical schools. Yet a sustained course of academic lectures seems to have been first offered only in 1851 by David Skae, physician of the Royal Lunatic Asylum in Edinburgh. Organicist Skae proposed that types of insanity be classified on the basis of the bodily illness causing them, hence “mania of masturbation,” “mania of pregnancy,” and so forth, ending up with a list of 25: “Each may be described as a disease presenting a certain variety and kind of mental symptoms,” he said.[55] In Dr. Skae’s classroom at the Edinburgh Asylum in Morningside, disease thinking ran riot.

In London, it was William Sankey who in the 1860s at University College

first began a course of psychiatry lectures. Sankey had been drawn to the subject of mental illness around the age of 40, having spent the first part of his career at the London Fever Hospital. Because many febrile illnesses cause delirium, the loss of orientation combined with psychotic symptoms, Sankey's interests in fever may have led him naturally toward psychiatry. A typical organicist, he did not doubt that melancholia possessed a morbid anatomy, though he was uncertain of its exact nature.[56] In 1854, he became the head of the female ward of the Middlesex County Asylum at Hanwell, leaving a decade later for a private clinic in the provinces, the Sandywell Park Lunatic Asylum in Gloucestershire. Around 1865, Sankey began as well to lecture on mental diseases at University College.[57]

Sankey got this post through the failed initiative of a young fellow of University College named Henry Maudsley. Maudsley, 30 at the time, had had a brilliant record as a medical student, graduating MB (medical bachelor, the British equivalent of the American MD) in 1856. At the surprising age of 24, he was appointed medical superintendent of the Manchester Royal Lunatic Hospital, coming down to London three years later to edit the flagship organ of British psychiatry, the *Journal of Mental Science.* In 1865, Maudsley proposed to University College that, as instruction in mental diseases was already available in Edinburgh, Paris, Vienna, and Berlin, a course be offered in London as well, because "every medical man has in his practice to deal with insanity as with any other disease; and it usually falls to him to deal with it at that early stage of the disease when there is always the best hope ... of effecting a cure." The appointments committee of the College was convinced by his logic, but passed Maudsley over in favor of Sankey for the post.[58]

Maudsley had circumstances going for him other than his brilliance. In 1866 he had married the youngest daughter of the famous John Conolly, the alienist who in 1850 had widely publicized the virtue of abolishing restraints, as his predecessors had done at Hanwell and elsewhere. [59] With one swoop, Maudsley had now behind him the authority of his father-in-law (who died a month after the wedding). Maudsley also took over the lease of a small private nervous clinic, Lawn House, that Conolly had established in his own home in Hanwell, with quarters for eight patients, all wealthy females. Lawn House, plus Maudsley's consulting practice at his other residence in Hanover-square in London, became the basis of a considerable fortune.[60]

In 1869 Maudsley was appointed professor of medical jurisprudence,

essentially a professorship in psychiatry, at University College Hospital. He went on to become the best-known psychiatrist of the Victorian era. He believed psychiatric illness to be a physical disorder of the body just as any other. “When a person is lunatic,” he said in 1870, “he is ... lunatic to his fingers’ ends.” For Maudsley as for Griesinger, mind disease was brain disease. “Mental disorders are neither more nor less than nervous diseases in which mental symptoms predominate.”[61] In the context of Victorian psychiatry there was nothing at all daring or iconoclastic about these theories. Maudsley articulated what virtually every psychiatrist and medical man of the day believed, but he did so from a position of great authority, which gave him an opportunity to try and show that he was right.

In 1907, Maudsley offered £30,000 to the London County Council, the body that administered the asylums of the metropolis, for a new asylum on conditions that would have been familiar to Griesinger: that it accept only recently ill people, that it make provision for teaching and research, and that it be in the heart of the city near the medical schools. Completed in 1915, the new building was first used as a military hospital and only in 1923—four years after Maudsley’s death—did it start to become the medical facility he had envisioned for “exact scientific research on the causes and pathology of mental diseases.”[62] England, therefore, had to wait almost 60 years before getting its first university psychiatric clinic on the German model made famous by Griesinger in 1865.

The Achilles’ heel, or genius if one prefers, of English psychiatry was that it was all clinical medicine and little science. The English were known as superb observers, clinical investigators, and examiners, but their clinical findings lacked the kind of anchor in the natural sciences at which the Germans were so gifted. Said a young Swiss medical student named Adolf Meyer after visiting London and Edinburgh in 1891, “One comes closest to the truth about English [sic] medicine in saying that it’s conceived as the art of healing, to which science is subordinated. Practical matters receive priority everywhere....”[63] It was not that the British were by nature more pragmatic or less inclined to rigor, merely that they were endowed with a system in which the teaching hospitals were dependent on charity. Government givers, as in Germany, had a penchant for science because its triumphs added to national prestige. Private givers, as in England, preferred to endow care because they were acting for humanitarian and dynastic reasons. Thus the kind of system that each country adopted made a big difference in the kind of psychiatry it

produced.

To understand the backward state of American psychiatry at the turn of the century one must bear in mind the inferior condition of American medicine in general. “In 1900,” writes historian Rosemary Stevens, “less than 10 percent of practicing physicians in the U.S.A. were graduates of genuine medical schools. About 20 percent had never attended medical school lectures. The majority were the products of apprenticeship or of the proprietary schools.”[64] It cannot surprise us therefore that research and education in psychiatry in the United States got off to a slow start. In 1868, the year of Griesinger’s manifesto, John Gray, the progressive superintendent at Utica State Hospital, persuaded the trustees to establish a pathology laboratory to make sections of the brain and spinal cord. Young Dr. Edward Hun of Albany, two years out of medical school at Columbia, was hired to do this work, and though we know little of his actual activity save that he left in 1873, we do know that he wrote on such topics as “the pulse of the insane” and “haematoma auris,” or blood blisters within the ear sometimes called “insane ear” and thought characteristic of madness (in fact they were the result of asylum attendants clubbing the patients).[65] One cannot imagine a figure of such meager stature at a comparable post in Europe.

Hun was followed at Utica by John Deecke, a supposed scientific stalwart who cut a considerable figure in the eyes of visiting foreign physicians. When, for example, English psychiatrist John Bucknill came over in 1876, he found that “Dr. Theodore [sic] Deecke devotes his time exclusively to pathological investigations; and is engaged at the present time in producing photographs of cerebral and spinal sections of wonderful size and accuracy.”[66] A scientific triumph, surely. Yet an observer closer to the scene tells us that Deecke, “who was a technician rather than a pathologist ... could make sections but could not interpret them.”[67] Deecke had never studied brain anatomy in medical school. The visiting British (unlike the eager young Dr. Farrar in London) did not discuss with Deecke what they were actually looking at. The “pathological laboratory” at Utica came to an end with Gray’s death in 1886 and the entire episode left “the tradition of the histopathologist of the Utica State Hospital” as something of a joke among American psychiatrists.[68]

At this point, the story shifts to the young Adolf Meyer, who in 1891, had been visiting European asylums. Just after earning his MD at Zurich in 1892, Meyer migrated to the United States, to study neurology, he said.[69] Meyer

was persuaded to make this ostensibly bizarre choice by a chance conversation in Berlin with an American neurobiologist who lauded the University of Chicago (perhaps Meyer believed possibilities for advancement were better in America than in Switzerland).[70] He landed in Chicago, which he found a scientific desert.[71] Then in 1893, Meyer was appointed staff pathologist at a mental hospital in Kankakee, Illinois, the Illinois Eastern Hospital for the Insane, which had opened its doors in 1879 as the first "noncongregate" asylum in the United States, meaning that the patients lived in rows of little stone houses rather than in vast barracks.[72] Progressive though Kankakee was architecturally, at the time of Meyer's arrival the staff was "hopelessly sunk into routine and perfectly satisfied with it." Meyer began by offering a course in neurology to them, then saw that it was not hitting home. "When I realized this, I started with something in which the physicians were more closely interested every day. I began by having the new patients given an examination in the presence of the whole staff, quite unofficially, during an hour usually spent in recreation."[73] This was Meyer's hallmark: extensive history-taking, extensive investigation, extensive notetaking. If the British fetishized nonrestraint and the Germans the microscope, Meyer fetishized "the facts." From this point on, Meyer would move as a kind of Johnny Appleseed through the landscape of American psychiatry, dropping the seeds of what he always conceived as general "medicine" wherever he went (he tended to use the terms medicine and psychiatry synonymously).

In 1895, Meyer left Kankakee for the asylum at Worcester, Massachusetts, an institution created in 1833 by the original reform wave in American psychiatry but that, by the turn of the century, had become a warehouse of the mad. Meyer had come to Worcester almost through happenstance: At a convention of psychiatrists in 1895, he had been discussing with Edward Cowles, director of the McLean Hospital in Boston, the need for a pathology laboratory in the public sector at Worcester similar to the laboratory that Cowles had already established at McLean. The director of Worcester then wrote Meyer offering him the post, assuring him that he would be absolutely free to undertake whatever he wished in terms of research and training and not just be limited to autopsies. Meyer at this point was growing tired of autopsies anyway, beginning to interest himself in "the living patient."[74] Yet the psychiatry that he brought to Worcester was still a resolutely biological one. "The fundamental principle on which the psychiatry of today is built,"

he wrote in 1897, “is undoubtedly this biological conception of man.” Exactly in line with his European preceptors, Meyer in his first decades in the United States saw the brain as the substrate of all mental activity: “We cannot conceive a disorder of the mind without a disorder of function of those cell mechanisms [of the brain].”[75]

After six years at Worcester, in 1901 Meyer was appointed director of the Pathological Institute in New York City, founded in 1895 to study brain sections from the state asylums. The Pathological Institute had drifted under its first director and now Meyer was called in to redirect it with the straightedge of German science.[76] In 1902, he relocated the Institute to one of the buildings—a disused bakehouse—of the Manhattan State Hospital for the Insane on Ward’s Island in New York.[77] Increasingly in these years, Meyer came to view the proper site for the study of psychiatry as the bedside rather than the “deadhouse.” Yet his interests remained emphatically biological rather than psychosocial. For example, in 1903 he called the asylum physicians of all the state hospitals together to talk about his new agenda. “I sketched the general principles of pathology as adapted to psychiatry, the methods of the physical examination of the insane, and the most frequent issues and difficulties of general physical and neurological diagnosis, and some of the latest observations in these directions, such as one-sided apraxia [inability to carry out intentional movements] and one-sided delirium.”[78]

When Meyer became professor of psychiatry at Johns Hopkins in 1910, he surveyed a landscape bereft of university science. Of the few laboratories approaching European standards, two he had created himself, at Kankakee and Worcester; several state hospitals had German-style laboratories, as for example Iowa and Minnesota; and the Government Hospital for the Insane (later St. Elizabeths Hospital) in Washington, opened in 1855, was building a scientific department. McLean Hospital probably approached the European model closest with its corps of physicians and small number of beds.[79] By the turn of the century, scarcely an American university was implicated in the systematic study of psychiatry, though many offered lecture courses.[80]

Characteristic of the development of psychiatry in the United States, therefore, was the separation of teaching and research, the opposite of the Continental model. This is why the founding of the Johns Hopkins Medical School in 1893 was of such significance: It promised to bring together science and the clinic, and Meyer’s journey to Baltimore in 1910 would accomplish this for psychiatry. It was, ironically, Meyer’s tenure at Hopkins

that put an end to the first biological psychiatry in the United States.

## *Degeneration*

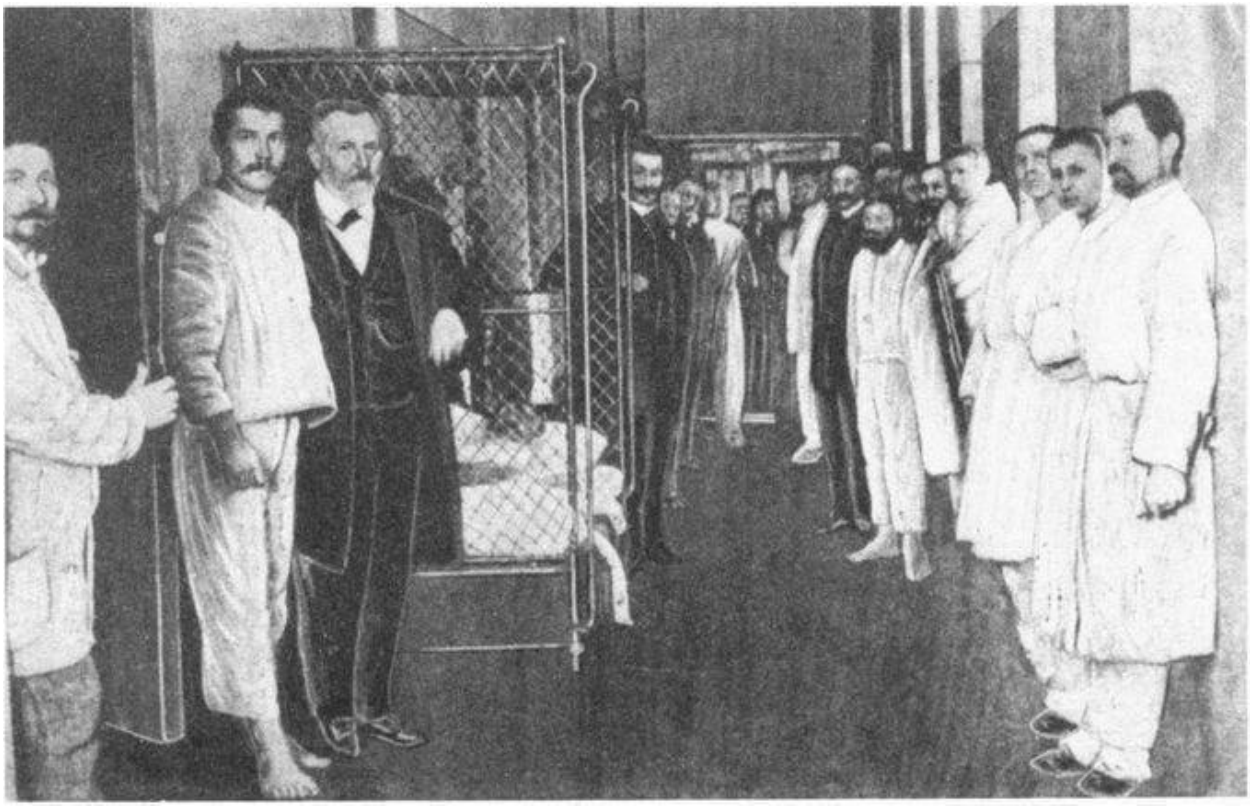
Psychiatrists of the nineteenth century pioneered the modern understanding of the genetics and biology of neuroscience. But then they went one step further. They asserted that not only do the major mental illnesses have a heavy biological and genetic component, but that these illnesses get worse as they are passed from generation to generation, causing progressive degeneration within family trees and within the population as a whole. Here psychiatry stood on very risky ground, ground that in fact would crumble into catastrophe as these ideas were appropriated by politicians as an agenda for action.

There is some truth to the concept of degeneration, although the term rings infamously on late-twentieth-century ears tutored to its misuse in the Holocaust: Some diseases with a psychiatric and neurological presentation do become worse as they are passed on. The genes that cause these illnesses expand in size (called technically trinucleotide repeat mutations) as the illnesses descend the family tree. The number of DNA base pairs becomes larger. Examples are the “fragile-X” syndrome, the second commonest cause of mental retardation after Down’s, and Huntington’s disease, involving involuntary, jerky movements (chorea) and mental deterioration. [81] So there is something to the notion of deterioration across the generations for select illnesses. Yet by the end of the nineteenth century, the advocates of degeneration had expanded the concept to embrace much of psychiatry and anthropology. The idea of degeneration began to bolster such social policies as sterilization, euthanasia, and persecution of the Jews on the grounds that they were a “degenerate” people.

In 1857, Benedict-Augustin Morel launched the concept of degeneration on its historical trajectory. Shocked by the “incessant progression” in Europe of such evils as general paralysis of the insane, epilepsy, suicide, and crime, Morel was attempting to identify underlying “natural forces” shaping the destiny of the human condition. He was struck that many of his patients looked funny. Some of the patients with mental retardation (“the cretins”), for example, had goiters (a result of iodine deficiency). Asylum patients,

generally speaking, seemed to Morel to have a “special cachet in their physiognomy.” What was the matter with these people? It was not just that they were suffering from inheritable insanity, he thought. All psychiatrists of Morel’s day believed mental illness to be directly inheritable. It was that “they recapitulate in their bodies the pathological organic characteristics of a number of previous generations,” that they were on the receiving end of the combined weight of generations of madness. Borrowing a term current in the comparative zoology of his day, Morel decided to call this pathological momentum within a family tree “degeneration.” And not only was it bad for the family, it was hurtful to society. Morel wrote, “The degenerate human being, if he is abandoned to himself, falls into a progressive degradation. He becomes ... not only incapable of forming part of the chain of transmission of progress in human society, he is the greatest obstacle to this progress through his contact with the healthy portion of the population.” Happily, “The span of his existence is limited as is that of all monstrosities.”[82]

Richard von Krafft-Ebing, professor of psychiatry in Vienna 1892-1902, on clinical rounds at the Vienna General Hospital. He is identified with “degeneration” theory and with his study of sex life *Psychopathia Sexualis* (1886) (courtesy of Institut fur Geschichte der Medizin, Vienna).



Thus Morelian degeneration was loosed upon the world. It might start as an acquired characteristic: alcoholism for example, or dissolution in the slums of the big cities. These vicious characteristics would then pass into the individual's seed and be transmitted via heredity, worsening with each generation. What began as maternal tuberculosis might end three or four generations later as dementia and sterility, he thought. This was truly a challenging agenda both for the social hygienists (clean up the slums and bars) and the healing professions, for if acquired characteristics could become inherited, action was required to remedy these evils. There was also another kind of action agenda. Morel believed, though he did not develop the point, that the degenerate should be sequestered.[83]

In every country, Morel had his signal advocates, representatives on mission to carry the doctrine of degeneration to the local psychiatric community. In Central Europe, it was Viennese psychiatry professor Richard von Krafft-Ebing, who is remembered by posterity mainly as the author of *Psychopathia Sexualis*, a kind of schoolboy's masturbatory compendium published in 1886 of what would later be called the alternative sexualities. Yet *Psychopathia Sexualis* merely gave Krafft-Ebing later notoriety. What

lent him his initial authority were the big textbooks on forensic psychiatry and general psychiatry he wrote in the 1870s.

Born and educated in Germany, in the late 1860s Krafft-Ebing was a staff psychiatrist at the Illenau asylum in Baden when he first warned readers of Morelian degeneration as a cause of criminality.[84] Later he declared, "Madness, when it finally breaks out, represents only the last link in the psychopathic chain of constitutional heredity, or degenerate heredity." [85] This was good Morelian doctrine, and its amplification in Central Europe increased as Krafft-Ebing himself climbed. In 1874, he became director of the provincial asylum in Graz, Austria, and professor of psychiatry at the university there. Five years later he wrote a textbook that became the German bible of degeneration theory. A sample: "[In degenerates] it is specially frequent for sexual functioning to be abnormal, in so far as there is either no sexual drive at all, or it is abnormally strong, manifesting itself explosively and seeking satisfaction impulsively, or abnormally early, stirring already in early childhood and leading to masturbation. Or it may appear perversely, meaning that the kind of satisfaction is not oriented to reproduction."[86] By the time he wrote *Psychopathia Sexualis* in 1886, Krafft-Ebing, shortly to be summoned to a chair in Vienna, was seeing degeneration literally under the bed. The onanists, homosexuals, and premature ejaculators who paraded through its pages (later editions became even more colorful) were virtually without exception stamped as "degenerates."[87] The book remains a classic example of psychiatry run off the rails, of the misuse of scientific authority to demonize cultural preferences. As one colleague said of Krafft-Ebing, "He was a man who was gifted in literary terms, yet scientifically and critically he was incapable to the point of feeblemindedness."[88]

In France, Valentin Magnan became the main standard-bearer of degeneration after Morel's death in 1873. Magnan first warmed to the subject in his lectures at Sainte-Anne in 1882, producing a decade later a widely read summary of the subject. Coming as he did after Darwin, Magnan interpreted the degenerates as losers in an epochal battle for survival among the species. A society that did not wish to succumb in the "hereditary struggle for life" would trim its load of degeneration, he said in 1895 with his collaborator Maurice Paul Legrain.[89] "Degeneracy is more than an individual disease, it is a social menace: It is important to combat it with a rigorous form of social hygiene. One must not forget that the degenerate is often a dangerous individual against whom society should and must reserve the right to defend

itself." Even better, the authors hinted ambiguously in the last sentence of their work that one might "cut off the problem at its roots."[90] Subsequent developments did not belie the ominousness of this promise.

Although it was Sankey who in 1857 introduced Morel's ideas to England, their primary apostle was Henry Maudsley.[91] As early as 1870, Maudsley suggested, with Morel, that his colleagues search for the stigmata of degeneracy in "malformations of the external ear ... tics, grimaces, or other spasmodic movements of muscles of face, eyelids or lips ... stammering and defects of pronunciation," as well as in eyes that possessed "a vacantly-abstracted, or half-fearful, half-suspicious, and distrustful look." "These marks are, I believe, the outward and visible signs of an inward and invisible peculiarity of cerebral organization."[92]

These teachings were absorbed by patients and physicians alike in late Victorian England.[93] How difficult it was, noted Samuel Strahan, senior medical officer of Berrywood Asylum in Northampton, to elicit a family history of mental illness from relatives who were all terrified at the specter of degeneration in the family tree. "We are all only too well acquainted with the manner in which these people, even in the poorer ranks of life, endeavour by every means to keep from us a knowledge of such family taint." The percentage of patients with family taint ranged in the statistics considerably. Yet in Strahan's view these statistics measured mainly "the amount of prevarication and untruth practised by the relatives of the insane." In reality the "inherited insane diathesis [tendency]" was to be found in "the majority of such cases," he said.[94] This is what most British alienists came to believe by the last two decades of the nineteenth century. At Brookwood asylum in Surrey, the percentage of patients in whom physicians whiffed a hereditary causation increased from 4 percent in 1870-1872 to 40 percent by 1890-1892.[95] Doubtless, many of these patients did have a positive family history but that is not the point. It is that many members of the psychiatric profession in England and elsewhere became convinced that a juggernaut of heredity born across the generations was bearing down on their patients, which is very different from noting that a genetic component may be present among many other circumstances in the unleashing of mental illness in a given individual.

Within psychiatry itself, the belief in degeneration in the sense of an inexorably gathering genetic fate, fell relatively quickly by the wayside. By the *belle époque,* degeneration had started to go out of style among alienists.

Magnan died in 1916, an isolated and disbelieved old man.[96] Samuel Strahan, the social Darwinist, left medicine and went into law. Degeneration started to become an object of ridicule among physicians. In a lampoon of 1911, Freud's collaborator Wilhelm Stekel makes the fictional Viennese coffeehouse "At the Sign of Degeneration" (*zur Degeneration* ) a rendezvous for doctors.[97]

A new generation of psychiatrists began to assess these doctrines with a cool eye. As Oswald Bumke, then a 31-year-old private docent at the psychiatric clinic in Freiburg, said in 1908: "The great edifice that the doctrine of inheritability had once erected has been demolished in recent years piece by piece, and all that remains is bits of rubble." His greatest problem at the moment was dissuading general practitioners from putting down "hereditarily burdened" *(erblich belastet)* as the cause of psychosis in their patients. Heredity does play a role, he told his target audience of family doctors, but let's not exaggerate it.[98] Other nonbiological interpretations, such as Karl Jaspers' 1913 "phenomenology" (stressing empathy with patients' subjective symptoms[99]), were gaining pride of place. Thus by the beginning of World War I, degeneration was being discredited within psychiatry. In the years between the World Wars, those who continued to preach that mental illness accelerated across the generations like an express train were viewed as marginal, eccentric figures.

But the genie was out of the bottle. In the last quarter of the nineteenth century, degeneration passed from an object of discussion within the closed world of academic psychiatry to the boulevard press. In a demonstration that ideas have consequences (and that physicians must be very careful about the correctness of what they believe before they make announcements to the public), the public seized with horror on the concept of degeneration. The educated middle classes began believing implicitly that European society was doomed unless action was taken to stem the poisoning of the well of heredity.

Novelist Émile Zola, in his series on the fortunes of the Rougon and Macquart families, depicted vast social forces battling each other, driven by the iron laws of social Darwinism; the degenerate were destined to slide under. As soon as we learn in Zola's 1885 novel Germinal that young Étienne Lantier's mother was a Macquart, we know that his fate is sealed:

"You shouldn't drink," Catherine says to Étienne.

"Oh, don't worry, I know I shouldn't."

> Étienne nodded his head; he hated liquor with the hatred of the last-born of a race of drunkards, the hatred of one who suffered so from a wild, alcohol-steeped inheritance that the least little drop was poison to him.

But Catherine has her own hereditary problems. As the wicked Chaval sets about to rape her, "She fell back on some old ropes and stopped resisting—submitting to the male, before her time, with that hereditary submission which made the girls of her race tumble in the open air while still in their childhood."[100] On and on it goes, the characters, all victims of degeneration, driven through life by their genes.

Part and parcel of European culture, the fateful notion of degeneration was picked up by the eugenists, by social-hygienists intent on combating mental retardation with sterilization, and by antidemocratic political forces with a deep hatred of "degenerate" groups such as homosexuals and Jews. Psychiatry's responsibility for all this is only a partial one. Academic psychiatrists in the 1920s were not generally associated with right-wing doctrines of racial hygiene, though there were exceptions to this, such as the Swiss psychiatrist Ernst Rüdin who after 1907 worked at the university psychiatric clinic in Munich, and the Freiburg professor Alfred Hoche who in 1920 coauthored a justification for euthanasia. [101] Academic medicine in Germany on the whole stood waist-deep in the Nazi sewer, and bears heavy responsibility for the disaster that followed. After 1933, degeneration became an official part of Nazi ideology. Hitler's machinery of death singled out Jews, people with mental retardation, and other supposedly biological degenerates for campaigns of destruction.[102]

The Nazi abuse of genetic concepts rendered any discussion of them inadmissible for many years after 1945. The notions of degeneration and inheritability became identical in the minds of the educated middle classes. Both were synonymous with Nazi evil. After World War II, any reference to the genetic transmission of psychiatric illness, whether as one factor among many or as inexorable degeneration, became taboo. The mere discussion of psychiatric genetics would, in civil middle-class dialogue, be ruled out of court for decades to come.

## *The End of the First Biological Psychiatry*

The first biological psychiatry as a clinical approach died long before the Nazis. It was not necessarily discredited by research findings. That's not the way paradigms change within medicine. People simply lost interest in brain anatomy once a new way of looking at psychiatric illness appeared on the horizon. The new approach saw illness vertically rather than cross-sectionally: trying to understand the patient's problems of a given moment in the context of his or her lifetime history, in contrast to the biological approach of trying to correlate the symptoms of the moment with neurological findings and with brain findings postmortem. Seminal in introducing this new vertical manner of looking at illness was Emil Kraepelin.

Kraepelin and colleagues. Emil Kraepelin (at lower right holding hat) and colleagues at a meeting of the Association of Southwest German Neurologists and Psychiatrists (Wanderversammlung der südwestdeutschen Neurologen und Irrenärzte), probably sometime between 1902 and 1904. Kraepelin was then professor of psychiatry at Heidelberg. From left to right in the top row are Albrecht Bethe (Strasbourg), and Aloys Alzheimer (Heidelberg). Franz Nissl (Heidelberg) seated to the left of Kraepelin cradling his characteristic walking stick in his arm. Robert Gaupp, Kraepelin's later opponent who was then at Heidelberg, is seated at lower left with hand on knee. The photograph was found in the papers of Dr. C. B. Farrar, courtesy of the estate. (Another copy of the photograph, with some incorrect identifying information, was published in the German edition of Emil Kraepelin's autobiography.)

It is Kraepelin, not Freud, who is the central figure in the history of psychiatry. Freud was a neurologist who did not see patients with psychotic illness. His doctrine of psychoanalysis, based on intuitive leaps of fantasy, did not stand the test of time. By contrast, Kraepelin and his data cards provided the single most significant insight that the late nineteenth and early twentieth centuries had to offer into major psychiatric illness: that there are several principal types, that they have very different courses, and that their nature may be appreciated through the systematic study of large numbers of cases. Ironically, Kraepelin, in his impatience with hypothetical theories about the brain, brought the first biological psychiatry to an end.

The American Kraepelin was Adolf Meyer, who helped to terminate the first biological psychiatry in the United States. Although Meyer never became an international figure (his brief vogue in Britain notwithstanding), he was the most prominent American psychiatrist before World War II, and his distinctive style did much to shape and misshape the evolution of American psychiatry in general. One is therefore entitled to speak of Kraepelin and Meyer in the same breath, keeping in mind that the influence

of the former was worldwide and enduring, and that of the latter evanescent and parochial—yet important in the American story.

Kraepelin's psychiatric career began in revolt against biological psychiatry. [103] Born in 1856 in North Germany, he earned his MD in 1878 at Würzburg, then went down to Munich to serve as a resident of the brain biologist Bernhard von Gudden. The clinic in Munich, founded in 1859, had figured among the early university psychiatric clinics, even before that of Griesinger. Gudden, the second chief of that clinic, belonged to the generation that believed the mysteries of psychiatry could be unfolded through the microscope. Many influential researchers, such as the Swiss August Forel, graduated from the Munich clinic.

As Kraepelin, a cool North German arrived in Munich, he became friendly with a young doctor from the Bavarian Palatinate named Franz Nissl, also a resident. But while Nissl, Gudden, and all the others were busy looking at brains through their microscopes, Kraepelin could not take part because he had an eye problem. Unlike the other Munichers, Kraepelin was also extremely interested in human psychology as a dimension of psychiatric illness. He had been keen on psychology since youth, and as a medical student avidly read the work of the experimental psychologist Wilhelm Wundt, who is considered the virtual founder of modern psychology. When Wundt established a psychological laboratory in 1879 at Leipzig, Kraepelin, the young psychiatrist, vowed to go there and study with him. Thus with Gudden's blessing, Kraepelin left Munich in 1882.

Kraepelin's attraction to psychology, and his repulsion at anatomically oriented psychiatry, were strengthened by his experiences in Leipzig. To earn his living, he had to work as an assistant with Paul Flechsig, whose only interest lay in the microscope and dissecting table. Kraepelin so hated Flechsig that he left his service after three months and plunged himself further into psychological studies with Wundt. In a story similar to that of Griesinger, who was young and inexperienced when he wrote a textbook, at 27 Kraepelin too attempted a textbook, a slim Compendium of no particular merit published in 1883.[104] He wrote it because he wanted to marry and needed the money.[105]

Kraepelin at this point was in almost the exact situation in which Freud found himself during these years: having to make a living because of a decision to marry. In 1884, Kraepelin became an asylum physician, just as Freud in 1886 entered private practice to earn a livelihood. But unlike Freud,

luck struck early for Kraepelin. In 1886, he received a professorship of psychiatry at what was then Dorpat University, after World War I called Tartu University, in the Estonian city of the same name.

At Dorpat, Kraepelin started to become intrigued with the kind of observation that had failed to interest the Gudden school: What happened to patients at the end of the road? How did their illnesses turn out? Yet his inability to speak Estonian left him unable to investigate the subject systematically.

In 1890, Kraepelin left marginal Dorpat for a professorship of psychiatry at Heidelberg, a university clinic at the epicenter of German academic life. At Heidelberg, moreover, there was no pressure on psychiatrists to do neurology, which was treated in a separate clinic. Here Kraepelin came into his own. He could now combine his interest in patients' illness trajectories with the psychological approach he personally favored. He could also incorporate in the Heidelberg clinic the anatomical psychiatry he had learned at Munich and which was still de rigueur for the day. Kraepelin undertook two innovations in particular. He began keeping little cards on each of his patients, noting previous history and condition at discharge. Second, in imitation of the comprehensive Griesinger-style department of psychiatry, Kraepelin wanted to surround himself with the best researchers available.

The people Kraepelin brought to the university psychiatric clinic in Heidelberg read like an honor roll of turn-of-the-century German neuroscience. In 1895, he lured his old Munich chum Franz Nissl to Heidelberg from the Frankfurt city asylum, which was then—under the dynamic Emil Sioli—something of a research center and not just a city lockup. Nissl was by now a famous neurohistologist who had discovered stains that permitted the nuclei and other structures of nerve cells to be seen, making possible the division of the cerebral cortex into numerous local areas and cell layers.[106] A curious man with a blunt demeanor, a large birthmark on his face, and an absolute indifference to sartorial display, Nissl was also well known among insiders for his bizarre work habits, spending the time from seven in the evening until dawn at his lab bench.[107] Working alongside Nissl at the Frankfurt asylum was another gifted young neurohistologist named Aloys Alzheimer, discoverer in 1906 of the disease later named after him.[108] Alzheimer and Nissl were great friends and notorious workaholics, and Alzheimer too was known for sleeping during the day and working through the night at his microscope. Alzheimer followed his friend to the Heidelberg

clinic in 1903. (The two were otherwise as unalike as cheese and chalk, Alzheimer an expansive family man, Nissl an obsessive-compulsive bachelor.) So on the organic side, the Heidelberg clinic had the two great students of the microscopic structure of the brain.

Although Kraepelin's interest in psychology was now waning, he nonetheless included measurement of patients' mental functions as part of their workup. He sought to engage bright young psychologically oriented physicians such as Willy Hellpach (who like Kraepelin had studied with Wundt[109]) to foster his larger plan of a comprehensive psychiatry ranging from the patients' dreams to their cerebral cortexes. Yet the emphasis was to be neither on psychology nor neuroanatomy but on the patient's course of illness across the years.

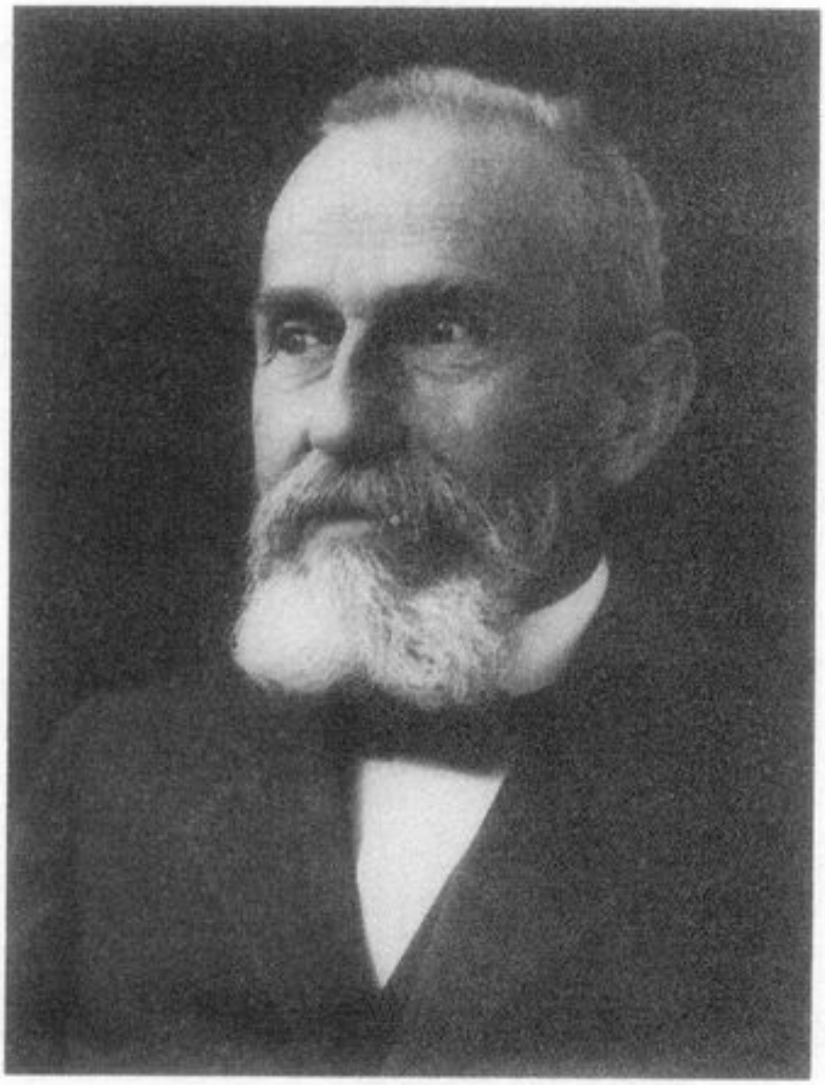
Looking at outcomes in psychiatric illness, and differentiating distinct diseases on the basis of those outcomes formed the essence of the Kraepelinian revolution. As the principal vehicle for disseminating his ideas, Kraepelin chose not scholarly articles but successive editions of the textbook he had begun in 1883. Starting with the third edition in 1893, new editions of this manual began to follow each other in close succession. So startling were the views Kraepelin began announcing that the entire psychiatric world would follow each new edition with rapt interest.[110]

Consider the state of psychiatric diagnosis before Kraepelin went to work. A half century of research in neuroanatomy and neuropathology had produced almost nothing in the way of concrete utility to clinical psychiatry beyond a picture of neurosyphilis. The biological psychiatrists had spawned a sprawl of clinical disease labels, each based on the particular circumstances associated with an illness ("masturbatory insanity," "wedding-night psychosis") or on the particular combination of symptoms ("chronic delusional disorder"), with almost no correlation to brain pathology. Multi-infarct dementia, neurosyphilis, thyroid deficiency, and the like were the only exceptions.

Yet before Kraepelin went to work, several fundamental issues had already been clarified. The notion of classifying diseases on the basis of course, or outcome, had already been raised by Karl Kahlbaum, one of the most neglected figures in the history of psychiatry. As a young physician at the Prussian state asylum at Allenberg, Kahlbaum had been irritated by Heinrich Neumann's assertion in 1859 that psychotic illness could not be subdivided into distinct diseases because there was only one "unitary psychosis"

(*Einheitspsychose*).[111] In 1863, Kahlbaum responded that indeed there were many distinct illnesses, one of which was psychosis in the young, which Kahlbaum called "hebephrenia."[112] Three years later, he left the public sector for the private, becoming assistant physician at a private nervous clinic in Görlitz. Taking over the clinic himself in 1867, Kahlbaum added a youth division, a "Medical Paedagogium," which gave him plenty of opportunity to see psychotic patients in their teens and early 20s. As Kahlbaum became the chief at Görlitz, he brought over from Allenberg a 24-year-old colleague, Ewald Hecker. The two worked together, and in 1871 Hecker wrote an article applying Kahlbaum's notion of hebephrenia to several clinical cases, describing in detail a malady specific to the young, resulting in mental disorientation and psychosis, and having a downhill course.[113] This was the first clinical description of schizophrenia with the claim that it represented a distinctive disease. Thus Kraepelin had Kahlbaum and Hecker to build on.

Eugen Bleuler, professor of psychiatry in Zurich 1898-1927, as seen around 1920. Bleuler coined the term "schizophrenia" in 1908 and was friendly for a while to Freud and the psychoanalytic movement (courtesy Psychiatrische Universitatsklinik Zurich).



He also had the work of two physicians at the Salpêtrière—Jean-Pierre Falret and Jules Baillarger—who had established in the early 1850s that mania and depression often occur not merely as isolated symptoms but in tandem with one another, alternating over a person's life course as "circular insanity."[114] (In 1899, Kraepelin would rebaptize this as "manic-depressive illness.") These French physicians had thus handed Kraepelin another building block, underpinning the notion that psychiatric illness consisted of an edifice of such building blocks, independent disease entities different in their symptoms and their outcomes.

Kraepelin and his residents would typically fill out a card for each patient and put them in the "diagnosis box." After they had had a chance to study the

patient, they would take the cards out again and enter on a list the patient's name and revised diagnosis. When the patient was discharged, the end result would be noted on the list. "In this manner we were able to get an overview and see which diagnoses had been incorrect and the reasons that had led us to this false conception."[115] On vacation, Kraepelin would take the lists and cards away with him and try to sort things out.

Only in the 1893 edition of his textbook did he discover that his file cards had started to make sense. In the preface, Kraepelin emphasized that all he knew he had learned from experience, that data gathered from patients lay behind every assertion, and that his ultimate objective was to try and cut nature at the joints: to identify natural disease entities. [116] Kraepelin's classification at this point was quite conventional, mostly concepts taken over from his predecessors. His typology did, however, contain one of the products of his card file: "psychic processes of degeneration," meaning illnesses that end in dementia. One subcategory of these was "premature dementia," or *dementia praecox*.

Kraepelin did not acknowledge, if he in fact knew, that Morel had first used the phrase dementia praecox in writings in 1852 and 1860.[117] But even before Morel—and later quite independently of him—other psychiatric writers had been talking about adolescent insanity. In 1873, Edinburgh psychiatrist Thomas Clouston described the "insanity of pubescence," which he also termed somewhat reluctantly the "hereditary insanity of adolescence."[118] In 1891, Clouston took another stab at the subject, calling it now "adolescent insanity and its secondary dementia." [119] Other physicians chimed in too. In August 1890 at a psychiatry meeting in Rouen, Parisian psychiatrist Albert Charpentier gave a paper on "les démences précoces," among which were included "les démences de la puberté."[120] By 1893, the notion of "démence précoce" was more or less in the psychiatric air.

Kraepelin's 1893 textbook was a historic document. He described in great detail what Hecker—whom Kraepelin believed to be the major previous authority—had only adumbrated in 1871. Kraepelin thought that the cause of dementia praecox was biological, using the term "psychopathic predisposition" (the 1890s equivalent of the 1990s euphemism "neurodevelopmental in origin").[121] The achievement of this 1893 work was formidable. In giving a careful account of dementia praecox, or schizophrenia, as a distinct disease, Kraepelin had handed psychiatry its most

powerful term of the twentieth century.

Surprises continued to unfold in later editions of this famous textbook. In 1896, in the reshuffling of file cards that produced the fifth edition, Kraepelin scrapped the part about degeneration. Dementia praecox was now a "metabolic disorder," set, to everyone's astonishment, next to thyroid psychosis and neurosyphilis. Yet what gripped his readers' attention most was his declaration in the preface that he was tired of grouping disorders according to their symptoms; he wanted to get to the inner nature of the illnesses, "as manifest in their course and outcome." "I have abandoned any effort to classify [psychosis] on the basis of the clinical presentation."[122] This was a decisive statement of Kraepelin's shift from studying the presumed causes of psychiatric illness, such as genetics, brain biology, and so forth, to concentrate on classifying illness in a way that would let one predict outcome. Prognosis, not cause, is the single most important word in understanding Kraepelin.

In this fifth edition in 1896, Kraepelin declared he was starting to put biological psychiatry behind him, deciding to concentrate instead on a psychiatry based on bedside observation of patients over time. He was not denying the validity of biological psychiatry, but simply declaring himself agnostic: "As long as we are unable clinically to group illnesses on the basis of cause, and to separate dissimilar causes, our views about etiology will necessarily remain unclear and contradictory."[123] In other words, cause is something that we, in our present state of knowledge, cannot know.

In the sixth edition of 1899, Kraepelin's ideas reached their definitive form, resulting in a classification of illness that provided the basis of the later *Diagnostic and Statistical Manual of Mental Disorders* of the American Psychiatric Association, the authoritative guide for world psychiatry in our own time (see pp. 565-580). Yet the point of all this classification, Kraepelin stressed, was not just ivory-tower pigeonholing but producing a psychiatry meaningful for patients and their families. Identifying natural disease entities should permit the physician to answer the question: Will my husband recover from his illness? As Kraepelin said in 1899, "The doctor's first task at the bedside is being able to form a judgment about the probable further course of the case. People always ask him this. The value of a diagnosis for the practical activity of the psychiatrist consists of letting him give a reliable look at the future."[124]

Kraepelin decided to divide all psychiatric illness into 13 big groups, most

of which were familiar (neurosis, febrile psychosis, mental retardation, and the like). Only two of the groups electrified his readers, and this is because Kraepelin had now split the vast world of those psychotic illnesses without an obvious organic cause into two neat camps: illnesses involving an affective component, and those without an affective component. (Affect means mood, whether the patient is depressed, manic, or anxious.) As he justified this dramatic compression of almost all affective illnesses into a single disease he said, "In the course of the years I have become more and more convinced that all [of the periodic and circular psychoses plus mania] are really just manifestations of a single disease process."[125] This process he labeled "manic-depressive psychosis" (*das manisch-depressive Irresein*).

This division of insanity into two big groups made diagnosis quite simple. If the patients were melancholic or euphoric, cried all the time, were always tired without a cause, or displayed any of the other signs of depression or mania, they were classed as "manic-depressive illness." If they were psychotic in the absence of an affective component, they had dementia praecox. If they were manic-depressive, they would probably get better; if they had what everybody was soon abbreviating as "d.p.," they probably wouldn't.

Thus by 1899, Kraepelin had elevated the two greater nonorganic ("functional") psychoses—manic—depressive illness and schizophrenia—to the top of the pyramid, where they remain in only slightly modified form to this day as the object of endeavor of serious psychiatry. "Old maid's insanity," "monomania," "moon madness," and the rest of it had been reduced to these two diseases, whose diagnosis could be made on the basis of the patient's history and current symptoms, and whose course could be predicted. The concept of prognosis was the vaulting stone of the entire arch: Patients with manic-depressive illness had a circular disorder that naturally would improve; patients with dementia praecox would deteriorate into what Kraepelin considered to be "dementia," or at least three-quarters of them would; a quarter of Kraepelin's dementia praecox patients recovered. Rather than reassuring patients and their families, Kraepelin's classification terrified them—unfairly so in retrospect. In fact, patients with schizophrenia do not become demented, they retain their intelligence, although their thought processes may become disorganized.

In recognition that these patients were not demented and that the onset was not necessarily early, in 1908 Zurich psychiatry professor Eugen Bleuler—

who considered himself to be a faithful student of Kraepelin's—proposed the term "schizophrenia" for what Kraepelin had been calling dementia praecox.[126] Yet at the meeting of the German Psychiatric Society at which Bleuler made this suggestion, and later in his 1911 book on "schizophrenia," Bleuler did shift the discussion somewhat from physical symptoms such as catatonia (rigid postures) to imputed psychological processes such as the French psychiatrist Pierre Janet's "splitting" of consciousness. [127] The term schizophrenia was probably an unfortunate choice, for subsequent generations of physicians and nonphysicians alike would associate it with some kind of splitting or divided consciousness. In schizophrenia, nothing is split. The disease is characterized by delusions, hallucinations, and disordered thought.

Yet these are lesser details. The main point is that the majesty of Kraepelin's overall structure transfixed the psychiatric world. For one thing, the interpretation of mental symptoms had been dethroned. The content of the psychosis no longer mattered. The precise symptoms the patient evidenced were unremarkable unless they gave evidence of the cause (neurosyphilis, thyroid disorder, etc.), or the course and outcome. Gasped Adolf Meyer in 1904, "The terms of a tradition of over 2,000 years are overthrown."[128]

In addition to providing a new way of classifying illness, Kraepelin's structure insisted that there were a number of discrete psychiatric illnesses, or diseases, each separate from the next. Depression, schizophrenia, and so forth were different just as mumps and pneumonia were different. Finally, being "Kraepelinian" meant that one operated within a "medical model" rather than a "biopsychosocial" model, as the battle lines later became drawn. A medically oriented psychiatrist believed in approaching psychiatric illness just as a cardiologist would approach heart disease, while remaining mindful that the psyche is subject to cultural shaping in ways that the heart is not. A psychiatrist attuned to Freudian or Meyerian biopsychosocial views might see illness more as the result of one's personal misadventures than of constitutional forces.

Kraepelin's doctrines were not received with complete approbation. Some psychiatrists refused to accept giving up syndromes (symptoms that seem to hang together reliably whatever the long-term outcome) and balked at subordinating all illnesses to these two huge categories. Alfred Hoche, the Freiburg psychiatrist who was proud of his own doctrine of supposedly immutable syndromes, sneered at Kraepelin's "annual opinions," a reference

to how the textbook kept changing.[129] Others resented Kraepelin's self-assured didacticism, the worldly Viennese psychiatrist Constantin von Economo calling him, "a North German village schoolmaster writ large."[130] The biological psychiatrists were unhappy at the dismissal of their doctrines: Wernicke scorned Kraepelin's work as "facile" (*feuilletonisch*).[131]

Internationally, Kraepelin's schema encountered greatest resistance among the French, who clung determinedly to Magnan's elaborate distinctions long after they had been abandoned elsewhere. There was a wonderful encounter in Paris as Nissl took the young American psychiatrist Clarence Farrar on an excursion to see Magnan's clinic. "Magnan," said Farrar much later, "had dropped everything to give us a field-day in his clinic.... The high point of the day was a brilliant, detailed demonstration of a very special case by Magnan himself."

Magnan demonstrated a patient as an "indubious example" of the chronic delusional disorder with downhill course (*délire chronique à évolution systématique*) that he himself had described. "Nissl listened with closest attention, now and then nodding appreciatively as the French-man made some fine psychological analysis of symptoms. The presentation complete, Magnan hopefully awaited Nissl's comment."

It was brief and direct: "A quite typical case of dementia praecox."

Farrar and Nissl learned the next day that after they had left, "Magnan had gone to his office, bowed his head over his desk, and wept."[132]

Kraepelin and his coworkers put an end to biological psychiatry as the dominant school, although individual representatives of this first biological school lingered on adjusting their microscopes into the 1920s. And it was not just because Kraepelin, agnostic about cause, had declared anatomy to be unimportant. It was because Kraepelin's microscopist colleagues themselves, such as Nissl, also helped to extinguish the notion that all psychiatric disorders had to have a basis in the brain. What Nissl and Alzheimer could find under their microscopes they declared "neurology." What they couldn't find was psychiatry. As Nissl said in 1908, "It was a bad mistake not to realize that the findings of brain anatomy bore no relationship to psychiatric findings, unless the relationships between brain anatomy and brain function were first clarified, and they certainly have not been up to the present."[133] In the second biological psychiatry, people would realize that Nissl had given away too much.

## *An American Postscript*

The story has an American postscript. Standing as a watershed in the United States was Adolf Meyer, a sort of American Kraepelin. Meyer's initial embrace of Kraepelin brought the doctrines of dementia praecox and manic-depressive illness to the United States, breaking with the old biological psychiatry. And Meyer's later rejection of Kraepelin helped plunge the United States into its psychoanalytic adventure.

Adolf Meyer, professor of psychiatry at Johns Hopkins University 1910-1941. The most prominent American psychiatrist of his day, he indiscriminately advocated all forms of psychiatric thinking from the biological to the psychoanalytic (courtesy of the Alan Mason Chesney Medical Archives of the Johns Hopkins Medical Institutions).

Meyer was not the only American psychiatrist to adopt Kraepelin's new system, as a whole squadron of young researchers regularly traveled back and forth from the Heidelberg clinic to their U.S. home bases. August Hoch, for example—who was originally Swiss but had a medical degree from the University of Maryland—had been hired in 1893 at the McLean Hospital in Belmont as a pathologist, a thoroughly biological type. After making several trips to Heidelberg, he started to lose interest (to the dismay of the McLean's superintendent) in laboratory studies and turned instead to "clinical" work,

the kind of thing Kraepelin had been doing. In the mid-1890s, Hoch persuaded the hospital to adopt the Kraepelinian system, resulting in a reclassification of all the records.[134]

In 1896, Meyer himself imported the Kraepelinian system to the Worcester asylum, where he had been hired as pathologist.[135] And though it was largely forgotten in the later celebration of Meyer as an enthusiast of psychoanalysis, in the beginning Meyer was a confirmed Kraepelinian, writing in 1897 of Kraepelin's "excellent work concerning clinical psychiatry." [136] He remained for years under Kraepelin's influence. When he took the professorship of psychiatry at Hopkins in 1910, it was to help shape a German-style psychiatric clinic that had already been planned, and that would open its doors in 1913. The clinic was an elegant fourstoried structure with windows guarded unobtrusively but not barred, a roof garden, and accommodations whose lavishness reminded more ribald staff members of "a general paralytic's dream."[137] There is no doubt that the model in Meyer's mind was Emil Kraepelin's brand-new psychiatric clinic in Munich (Kraepelin had gone from Heidelberg to Munich in 1904). The Phipps clinic, with its comprehensive facilities for both the psychological and the laboratory investigation of patients, had been very much cast in the Kraepelinian model.

But Meyer went further. He insisted on the integration of psychiatry into the rest of a general hospital, and treated psychiatry patients as "medical" in a way that went beyond even the Germans. Patients arrived at the Phipps clinic in an ambulance, for example, not in handcuffs, and usually were not under a certificate.[138]

It is puzzling, therefore, that Meyer later acquired the reputation of having always been anti-Kraepelinian.[139] One explanation is that his earlier writings had simply been forgotten, the bulk of his collected works appearing only after 1950, the year of his death. Certainly psychiatry's collective memory has blacked out that Meyer, the most honored American psychiatrist of the twentieth century, was once an ardent believer in degeneration. In 1895, for example, he called the ear a guide to nature's ample variety of types of "degeneration."[140] Meyer was even upset that Kraepelin had thrown out the notion of degeneration![141] Perhaps there was some amnesia.

An even greater obstacle to situating Meyer was Meyer himself. A second-rate thinker and a verbose writer, he was never, in his own mind, able to disentangle schools that were absolutely incompatible, and ended up

embracing whatever new came along. Thus Meyer fervently saluted the maniacal Trenton psychiatrist Henry Cotton, who believed in pulling out his patients' teeth and removing their large bowels to cure psychiatric illness caused, as Cotton believed, by "autointoxication." Meyer urged that this work, the "remarkable achievement of the pioneer spirit," be continued after Cotton's death in 1933.[142]

Yet later in life, Meyer did reject Kraepelin. He also put biological psychiatry behind him, which is not at all the same thing. Meyer ventured forth his own psychosocial models of mental illness, and turned his back on Kraepelinian classification in favor of his own bizarre terminology of "reaction types," employing such neologisms as "ergasiology," "pathergasias," and "kakergasias."[143] (The notion of reaction types itself came directly from such German anti-Kraepelinians as Ludwig Binswanger.) As for the causation of psychiatric illness, Meyer ended up adopting the position that everything is very complex.[144] Although true at one level, this kind of throwing up of the hands in despair to concentrate on collecting even more "facts" (Meyer's wont) is poisonous to the advance of a scientific discipline.

For better or worse, Meyer's soaring reputation brought the advance of the first biological psychiatry in the United States to an end. By the 1940s, he would be calling schizophrenia "psychogenic" and urging psychotherapy as the treatment of choice.[145] On both sides of the Atlantic, looking for the causes of mental illness in the biology of the brain had become a thing of the past.